

38.pdf

 University of Engineering & Management

Document Details

Submission ID

trn:oid:::3618:103124111

Submission Date

Jul 1, 2025, 2:33 PM GMT+5:30

Download Date

Jul 5, 2025, 12:34 AM GMT+5:30

File Name

38.pdf

File Size

1.1 MB

11 Pages

8,551 Words

53,930 Characters

26% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

-  **149** Not Cited or Quoted 20%
Matches with neither in-text citation nor quotation marks
-  **1** Missing Quotations 0%
Matches that are still very similar to source material
-  **17** Missing Citation 5%
Matches that have quotation marks, but no in-text citation
-  **0** Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

- 20%  Internet sources
- 21%  Publications
- 16%  Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Match Groups

- 149** Not Cited or Quoted 20%
Matches with neither in-text citation nor quotation marks
- 1** Missing Quotations 0%
Matches that are still very similar to source material
- 17** Missing Citation 5%
Matches that have quotation marks, but no in-text citation
- 0** Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

- 20% Internet sources
- 21% Publications
- 16% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Internet	dissem.in	2%
2	Internet	scholar.archive.org	<1%
3	Internet	www.mdedge9-ma1.mdedge.com	<1%
4	Internet	rcastoragev2.blob.core.windows.net	<1%
5	Internet	www.mdpi.com	<1%
6	Internet	arxiv.org	<1%
7	Internet	onlinelibrary.wiley.com	<1%
8	Internet	pubs.rsna.org	<1%
9	Internet	www.researchsquare.com	<1%
10	Internet	sibet.cas.cn	<1%

11	Internet	doczz.net	<1%
12	Internet	xuebao.sysu.edu.cn	<1%
13	Submitted works	University College London on 2025-04-09	<1%
14	Internet	fastercapital.com	<1%
15	Publication	Abdulrahman Alahmadi. "Towards ovarian cancer diagnostics: A vision transform...	<1%
16	Internet	link.springer.com	<1%
17	Submitted works	Addis Ababa University on 2025-06-16	<1%
18	Internet	ovarianresearch.biomedcentral.com	<1%
19	Publication	Namani Deepika Rani, Dr. Mahesh Babu. "Improved Rank-based recursive feature...	<1%
20	Internet	acsearch.acr.org	<1%
21	Publication	Dongmei Zhou, Jing Zhang, Jie Ma, Xiaowei Xi, Rui Ma. "Optimizing Deep Learning...	<1%
22	Publication	Gengshen Bai, Shiwen Huo, Guangcai Wang, Shijia Tian. "Artificial intelligence ra...	<1%
23	Internet	www.hindawi.com	<1%
24	Internet	www.researchgate.net	<1%

25	Internet	www.rpmj.ru	<1%
26	Internet	obgyn.onlinelibrary.wiley.com	<1%
27	Internet	www.acarindex.com	<1%
28	Internet	www.ijcmsr.com	<1%
29	Publication	P Dss Lakshmi Kumari, P. Maragathavalli. "Integrating Machine Learning Algorith...	<1%
30	Submitted works	University of Cumbria on 2023-04-18	<1%
31	Internet	ideas.repec.org	<1%
32	Internet	www.bspublications.com	<1%
33	Internet	www.medrxiv.org	<1%
34	Submitted works	Firat Üniversitesi on 2025-06-04	<1%
35	Internet	www.ncbi.nlm.nih.gov	<1%
36	Internet	www.nitrc.org	<1%
37	Internet	www.springermedizin.de	<1%
38	Publication	M. Kishi, D. Kawahara, R. Nishioka, I. Koh, K. Tomono, Y. Sato, I. Nishibuchi, Y. Mu...	<1%

39	Internet	dblp.dagstuhl.de	<1%
40	Publication	"Proceedings of International Conference on Machine Intelligence and Data Scie...	<1%
41	Publication	Rajwinder Singh, Hardeep Kaur, Jyoteesh Malhotra. "Optimizing ovarian cancer s...	<1%
42	Publication	Ranjan Parekh. "Principles of Multimedia", CRC Press, 2025	<1%
43	Internet	m.amedeo.com	<1%
44	Internet	techrights.org	<1%
45	Internet	www.peeref.com	<1%
46	Publication	Abdominal Imaging, 2013.	<1%
47	Publication	"Computing and Machine Learning", Springer Science and Business Media LLC, 20...	<1%
48	Publication	"ICCKE 2023 Subject Index Page", 2023 13th International Conference on Comput...	<1%
49	Publication	"Software Engineering Perspectives in Intelligent Systems", Springer Science and ...	<1%
50	Submitted works	Heriot-Watt University on 2025-04-10	<1%
51	Publication	Keerthivasan Chandradass, Ajin R Nair, Harikumar Rajaguru. "Early Detection an...	<1%
52	Internet	doaj.org	<1%

53	Internet	embc.embs.org	<1%
54	Internet	publishing.emanresearch.org	<1%
55	Internet	www.jetir.org	<1%
56	Publication	Reynolds, E.A.. "A Review and an Update on the Screening of Epithelial Ovarian Ca...	<1%
57	Submitted works	Australian National University on 2023-05-29	<1%
58	Publication	Ovarian Neoplasm Imaging, 2013.	<1%
59	Internet	www.coursehero.com	<1%
60	Internet	www.rapidinnovation.io	<1%
61	Publication	"ECR 2019: Book of Abstracts", Insights into Imaging, 2019	<1%
62	Internet	academictree.org	<1%
63	Internet	icimatingsociety.org.uk	<1%
64	Internet	ojs.revistacontemporanea.com	<1%
65	Internet	ouci.dntb.gov.ua	<1%
66	Internet	research.tees.ac.uk	<1%

67	Publication	Abhijit Bhowmick, Yogesh Kumar Choukiker, Indrasen Singh, Arumugam Nallana...	<1%
68	Submitted works	University of East London on 2024-06-24	<1%
69	Submitted works	University of Hong Kong on 2019-06-21	<1%
70	Internet	harbinengineeringjournal.com	<1%
71	Internet	www.dovepress.com	<1%
72	Publication	Chowdhury, Areefin-Ul-Hassan. "Mitigating the Greenhouse Gas Footprint of Hyd...	<1%
73	Submitted works	University of East London on 2025-04-16	<1%
74	Submitted works	University of Sheffield on 2021-09-22	<1%
75	Internet	iqjmc.uobaghdad.edu.iq	<1%
76	Internet	mmrjournal.biomedcentral.com	<1%
77	Internet	wjgnet.com	<1%
78	Internet	www.marsharivkin.org	<1%
79	Internet	www.nature.com	<1%
80	Internet	www.wjgnet.com	<1%

81	Submitted works	Bahcesehir University on 2025-05-05	<1%
82	Publication	Kanimozhi S, Sivakumar Rajagopal, Ananthakrishna Chintanpalli. "Recent Advanc...	<1%
83	Publication	L. Qi, X. Li, Y. Yang, M. Zhao, A. Lin, L. Ma. "Accuracy of machine learning in the pr...	<1%
84	Publication	Samaira Mehta. "OVision A raspberry Pi powered portable low cost medical devic...	<1%
85	Submitted works	University College London on 2025-04-01	<1%
86	Submitted works	University of Dundee on 2025-05-31	<1%
87	Submitted works	University of Leeds on 2023-03-23	<1%
88	Submitted works	University of Leicester on 2013-09-13	<1%
89	Submitted works	WorldQuant University on 2025-06-29	<1%
90	Internet	biosignaling.biomedcentral.com	<1%
91	Internet	data-science.meduniwien.ac.at	<1%
92	Internet	econpapers.repec.org	<1%
93	Internet	utsouthwestern.elsevierpure.com	<1%
94	Internet	www.frontiersin.org	<1%

95	Internet	www.polradiol.com	<1%
96	Internet	www.prelekara.sk	<1%
97	Internet	www.science.gov	<1%
98	Internet	www.thieme-connect.com	<1%
99	Publication	Eman I. Abd El-Latif, Mohamed El-dosuky, Ashraf Darwish, Aboul Ella Hassanien. "...	<1%
100	Publication	Evis Sala. "The role of functional MRI and PET/CT in evaluation of patients with pri...	<1%
101	Publication	MRI of the Female and Male Pelvis, 2015.	<1%
102	Publication	Mohamed El-khatib, Dan Popescu, Oana Teodor, Loretta Ichim. "Intelligent syste...	<1%
103	Publication	Smith, Lauren M.. "MRI of Structural & Functional Changes Associated with Weste...	<1%
104	Publication	Thien B. Nguyen-Tat, Anh T. Vu-Xuan, Vuong M. Ngo. "Design of a novel fuzzy ens...	<1%
105	Submitted works	University of Cumbria on 2025-02-28	<1%
106	Submitted works	University of Hertfordshire on 2019-02-28	<1%
107	Publication	V Sathiyavani, Dr. Vinothkumar Arumugam, Dr. M Anand. "Design and Developm...	<1%
108	Internet	asianresassoc.org	<1%

109	Internet	bcn.iums.ac.ir	<1%
110	Internet	id.scribd.com	<1%
111	Internet	journal.aastu.edu.et	<1%
112	Internet	pmc.ncbi.nlm.nih.gov	<1%
113	Internet	pubmed.ncbi.nlm.nih.gov	<1%
114	Internet	www.restack.io	<1%
115	Publication	Axel Geysels, Giulia Garofalo, Stefan Timmerman, Lasai Barreñada et al. "Artificial...	<1%
116	Publication	David J Collins. "Technology Insight: water diffusion MRI—a potential new biomar...	<1%
117	Submitted works	Deenbandhu Chhotu Ram University of Science and Technology on 2023-07-04	<1%
118	Publication	Functional Imaging in Oncology, 2014.	<1%
119	Publication	Meng-Lin Huang, Jing Ren, Zheng-Yu Jin, Xin-Yu Liu, Yong-Lan He, Yuan Li, Hua-Da...	<1%
120	Publication	S J K Jagadeesh Kumar, G. Prabu Kanna, D. Prem Raja, Yogesh Kumar. "A Compreh...	<1%
121	Submitted works	universititeknologimara on 2025-06-20	<1%
122	Publication	"Scientific and Educational Exhibits", European Radiology, 2009	<1%

123	Publication	Junming Jian, Yong'ai Li, Wei Xia, Zhang He et al. " Multiple Instance Convolution...	<1%
124	Publication	Mrutyunjaya Panda, Ajith Abraham, Biju Gopi, Reuel Ajith. "Computational Intelli...	<1%
125	Publication	Samridhi Singh, Malti Kumari Maurya, Nagendra Pratap Singh, Rajeev Kumar. "Su...	<1%
126	Submitted works	University of Nottingham on 2022-03-24	<1%

A Review of Deep Learning Techniques for Ovarian Cancer MRI Detection and Classification

K Vidhya

Computer Science and Engineering Department
Karunya Institute of Science and Technology
Comibator

Minu Cherian

Computer Science and Engineering Department
Karunya Institute of Science and Technology
Comibator

Abstract—High rates for ovarian cancer remain around the globe, as it is often diagnosed at a later stage, and there is limited screening. Detecting and classifying diseases such as cancers from medical images has emerged with remarkable success in the last few years with the advancements of deep learning (DL). This article addresses the deep learning applications in cancer detection and classification in the ovaries using magnetic resonance imaging (MRI). In this article, we provide a systematic review of state-of-the-art techniques, architectures, datasets, performance, and challenges of the area. It also proposes current and new techniques and advancements of profound learning frameworks for ovarian malignant growth conclusion and arrangements to existing research holes. As such, there is great strides in the field, although relevant roadblocks are limited datasets, lack of transparency of the models and clinical adoption. This review serves as a useful reference for researchers and clinicians exploring the integration of artificial intelligence in gynecologic cancer care.

Index Terms—Ovarian cancer, deep learning, magnetic resonance imaging, computer-aided diagnosis, medical image analysis, convolutional neural networks.

I. INTRODUCTION

A. Ovarian Cancer: A Challenge for Clinical Medicine

Ovarian cancer constitutes a significant global health challenge, being the seventh most prevalent malignancy among women worldwide and the deadliest gynecological cancer [1]. According to recent GLOBOCAN statistics, approximately 313,959 new cases of ovarian cancer are identified annually, leading to an estimated 207,252 fatalities [1]. This disproportionately elevated mortality rate can be primarily attributed to the advanced stage at which the disease is typically diagnosed; more than 70% of instances are identified at either stage III or IV, at which point the five-year survival rate dramatically declines to below 30% [2], [3]. In stark contrast, when identified in the initial stage, the five-year survival rate exceeds 90% [4], thereby highlighting the critical necessity for early detection.

Several factors contribute to the difficulties associated with early diagnosis. Primarily, the deep anatomical positioning of the ovaries within the pelvic cavity renders them inaccessible to direct physical examination. Furthermore, early-stage ovarian cancer frequently manifests with ambiguous, nonspecific symptoms that may easily be misattributed to more prevalent benign conditions such as abdominal distension, pelvic discomfort, early satiety, or increased urinary frequency

[5]. Moreover, the current diagnostic criteria for ovarian cancer exhibit insufficient sensitivity and specificity for widespread population use, as corroborated by comprehensive clinical studies such as the UKCTOCS (the United Kingdom Collaborative Trial of Ovarian Cancer Screening) [6] and the PLCO trial (Prostate, Lung, Colorectal, and Ovarian Cancer Screening Trial) [7].

The clinical challenges associated with ovarian cancer underscore the pressing necessity for advancements in diagnostic methodologies that can more accurately identify the disease at stages where it is amenable to treatment and effectively differentiate between benign, borderline, and malignant lesions.

B. Magnetic Resonance Imaging in Ovarian Cancer Evaluation

Magnetic Resonance Imaging (MRI) has emerged as a critical imaging technique for the evaluation of ovarian cancer, owing to numerous intrinsic advantages. In contrast to computed tomography (CT), MRI does not utilize ionizing radiation, thereby rendering it safer for continuous monitoring and for application in younger demographics [8]. MRI provides superior soft-tissue contrast resolution relative to CT or ultrasound, facilitating enhanced characterization of tissue constituents within ovarian masses [9]. Moreover, the multiplanar capabilities of MRI enable a thorough assessment of complex pelvic anatomy and an accurate determination of local disease progression [10].

Standard MRI protocols employed in the evaluation of ovarian masses generally encompass:

- T1-weighted imaging: Particularly advantageous for identifying hemorrhagic components, fat content, and solid enhancing regions subsequent to intravenous contrast administration.
- T2-weighted imaging: Crucial for differentiating cystic from solid elements and for characterizing the internal architecture of complex lesions.
- Diffusion weighted imaging (DWI): In order to assess tissue cellularity and the integrity of cellular membranes, diffusion weighted imaging (DWI) makes it easier to identify the random Brownian motion of water molecules. Malignant pathology is usually associated with restricted diffusion, which is defined by high signal intensity on DWI and low apparent diffusion coefficient values.

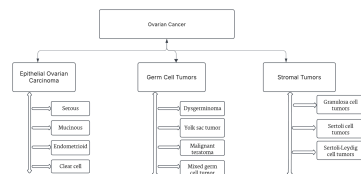


Fig. 1. Brief about Ovarian Cancer.

- **Dynamic contrast-enhanced (DCE) imaging:** Enables the analysis of perfusion patterns and contrast enhancement, with malignancies frequently exhibiting rapid enhancement followed by prompt washout [12].

The integration of these sequences provides complementary information that has improved the ability of radiologists to differentiate benign from malignant ovarian masses. Despite these advances, interpretation of ovarian MRI remains challenging and subject to inter-observer variability. The wide spectrum of imaging appearances of ovarian pathologies, overlap in imaging features between different entities, and subtlety of early-stage disease all contribute to diagnostic uncertainty. These challenges highlight the potential benefit of computer-aided diagnosis systems that can objectively analyze the complex multiparametric information contained within MRI examinations.

II. MRI-BASED CLASSIFICATION OF OVARIAN CANCER

Deep learning MRI-based classification of ovarian cancer: The most crucial steps The process starts with collecting data where a big dataset of MRI images of ovarian tumors is obtained. The data set must contain a variety of cases as well, such as benign, borderline, and malignant tumors, allowing the deep learning model to learn to differentiate them. A vital piece of this step is data annotation, wherein skilled radiologists tag the MRI scans to create the ground truth for supervised learning.

This can range from image enhancements to standardisation of inputs (to deep learning models), image preprocessing is performed once the dataset is available. Some common preprocessing steps involve noise removal using techniques like Gaussian or median filtering, contrast enhancement via histogram equalization or adaptive contrast stretching, and normalization to maintain a consistent intensity across images. Furthermore, resizing and formatting of MRI images can be performed to ensure they match the input size required by the deep learning model.

After processing, ROI extraction focuses on the ovarian region and removes unwanted structures. This is a critical step towards improving the accuracy of classification since it prevents the model to learn features correlated with unnecessary anatomical structures instead of ovarian cancer. Techniques for segmentation (such as U-Net, Mask R-CNN or other CNN based models) can be used to automatically separate the ovarian tumor areas from the background.

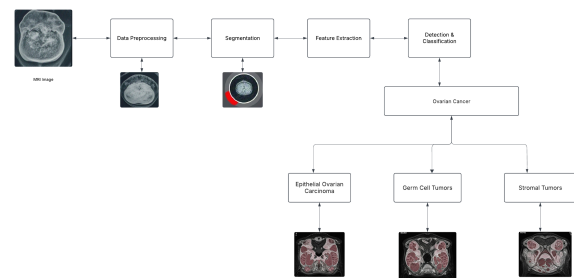


Fig. 2. Steps of ovarian cancer detection and classification involves several key steps to ensure high-quality input for analysis. The figure illustrates the following stages: (a) Data Preprocessing, (b) Segmentation, (c) Feature extraction, (d) Detection & Classification.

Following the extraction of the Region-of-Interest (ROI), deep learning architectures like Convolutional Neural Networks (CNNs) are used to extract features. CNNs are well known for extracting spatial and hierarchical features from images. Deep learning models automatically learn the input features rather than using handcrafted feature extraction as shortly and identify patterns such as texture, shape, and intensity variations that differentiate cancerous from non-cancerous tissues. Using transfer learning with pretrained models like ResNet, VGG or EfficientNet, a model can be trained on a large dataset like ImageNet and then fine-tuned on the MRI dataset as features.

After relevant features been extracted, the classification is carried by a deep learning model which has been trained to classify different ovarian cancer types. Most commonly classification, it can use models such as ResNet, VGG, DenseNet, or hybrid CNN-RNN architectures for this stage. The model learns from labeled MRI images and is trained nine by an appropriate loss function (for example, categorical cross-entropy when classifying multiple classes, and binary cross-entropy when identifying benign versus malignant tumors). Resize images, add padding, flip images, rotate, and add noise for data augmentation, dropout, L1/L2 regularization, and batch normalization are also used to make deep neural networks more robust for better class performance.

The model is then evaluated on its performance using several metrics such as accuracy, precision, recall, F1-score, and area under the ROC curve (AUC-ROC). These numbers are good at measuring generalization to unseen data. Cross-validation is a technique used to ensure the robustness of your model by splitting the dataset into training and validation subsets, multiple times, to avoid overfitting.

Finally, fine-tuning and optimization techniques are employed to improve the classification accuracy. Adjustments are made for hyperparameters such as learning rate, batch size, network depth, etc., to achieve best model performance. Furthermore, using the approach of transfer learning by using pre-trained deep learning architectures and customizing them to the context of the ovarian cancer MRI dataset can also be an alternative approach.

After you have a robust model, it will be deployed in a

clinical scenario where it will act as aid for radiologists in detecting and diagnosing ovarian cancer faster. This model can be embedded in a Computer-aided diagnosis (CAD) system, which allows automatic secondary opinions, which can avoid diagnostic errors. Future evolutions may further encompass federated learning for collaboration among multiple centers, explainable AI (XAI) for interpretability of models, and real-time MRI analysis for timely decision-making in clinical settings.

III. DATA PREPROCESSING

Preprocessing is a key aspect of examining medical imaging data such as quantitative multiparametric imaging of ovarian cancer, and It consists of various techniques used to improve data quality, eliminate noise, and facilitate the data for appropriate analysis. A detailed description of preprocessing steps, with supporting research results, is presented below.

A. Data Cleaning

Data cleaning is the very first and the most essential step in the preprocessing. This is a process of checking and correcting the errors in the dataset, like missing value detection, outlier treatment, and duplicate removal. Data Missingness in Ovarian Cancer Datasets Ovarian cancer datasets may exhibit missingness due to patient chart not being completed or due to technical reasons. Outliers can interfere with that analysis, making the results less accurate.

- **MissingValues:** The most common approaches used to account for missing values, are mean or median imputation, and more sophisticated techniques like multiple imputation by chained equations (MICE) [14].
- **Outlier Detection and Removal:** Outliers can be detected using techniques like Z-score or Isolation Forest algorithms. These are recognized, after which they are discarded or imputed [14].

B. Feature Selection

Feature selection is a preprocessing task aimed at reducing dimensionality by selecting only the relevant features. Maintaining predictive, cross-validated reproducibility, especially in the light of accumulation of potentially redundant features, is an essential concern in MRI analyses, as consecutive MRI sequences and multi-parameter representations lead to the Arnold class features (paradigm) of substantial dimension.

- **Recursive Feature Elimination (RFE) :** Paper proposes this technique which iteratively removes the least important features until the desired number of features is reached. It subsequently improved the accuracy of classification models for ovarian cancer datasets [14].
- **Feature Importance:** is used to assign a score to each feature based on how much it contributes to the performance of the model. The features with higher scores are selected for analysis [14].

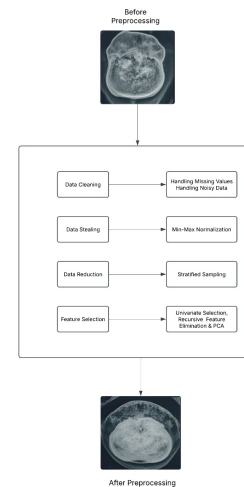


Fig. 3. Data preprocessing of MRI images of ovarian cancer involves several key steps to ensure high-quality input for analysis. The figure illustrates the following stages: (a) Data Cleaning, (b) Data Stealing, (c) Data Reduction, (d) Feature selection

C. Normalization

Normalization Framing Context Normalization is applied to convert the data into a common range, generally between 0 and 1, so that one or more features with a relatively larger magnitude do not overweight in our analysis and statistics. This is an essential step in the data processing of MR that is because intensity values may vary widely.

- **Min-Max Scaling:** This method scales the data to a fixed range, commonly 0 and 1, according to the minimum and maximum of the dataset [14].
- **Z-Score Standardization:** This method involves calculating how far each data point is from the mean in terms of standard deviations and scaling accordingly, yielding a new data set with a mean of 0 and a standard deviation of 1 [14].

D. Dimensionality Reduction

Dimensionality reduction methods aid in decreasing the number of variables in a dataset while maintaining the majority of the vital data. This is an important step for improving the efficiency of machine learning models and lowering computational costs.

- **PCA: Principal Component Analysis (PCA)** is a dimensionality reduction technique that converts the dataset into principal components which is basically a linear combination of original features. It is often used in MRI data analysis to decrease data dimensionality without losing significant patterns [14].
- **t-Distributed Stochastic Neighbor Embedding (t-SNE):** It is primarily used for the visualisation of high dimensional data into a lower-dimensional space. It is especially useful for revealing latent structures in MRI datasets [14].

E. Data Transformation

Data transformation means transforming data into a more useful format to analyze it effectively. This step is necessary for MRI where intensity values can be non-linear and/or skewed.

- Log Transformation: Logarithmic transformation helps to stabilize variance and produce a more normally distributed dataset, which is especially effective for highly skewed datasets [14].
- Square Root Transformation: As with the above test, this technique reduces the effect of extreme values and helps achieve symmetrical data [14].

Authors	Methods Used
Roopashri Shetty et al	Data cleaning, normalization
M. Sadeghi et al	Deep learning techniques
Jossy P. George et al	Convolutional Neural Network (CNN)
S. Banumathi et al	Data augmentation
Jori S. Carter et al	Pharmacokinetic modeling

TABLE I

DIFFERENT PREPROCESSING METHODS.

F. Feature Engineering

The process of creating new features from the existing data that help to improve the performance of the model. This is a critical process in MRI analysis, where raw data may be insufficient to depict necessary patterns.

- TextureBased Classifier Approach: The Texture-based classifier uses features such as Contrast, Correlation, and entropy from these images, which describe the texture of MRI images. Such characteristics can be highly informative to discriminate benign from malignant tumors [15].
- Shape and Size Information features about tumors, for example area, perimeter and circularity are derived characteristics that provide further insights about tumors helpful in classifying and diagnosis of tumors [15].

G. Data Augmentation

Data augmentation is a methodology that creates more data points by mixing up the existing ones to generate new samples. This cannot be understated, especially in MRI analysis, where the amount of available data may be limited.

- Rotation and Flipping: This technique includes rotation and flipping of MRI images to produce different transformations to augment the heterogeneity of the dataset and improve the model generalization capabilities [14].
- Noise Injection: This approach involves adding random noise to the MRI images, simulating real-world variances, and helping the model become more robust to various imaging conditions [14].

H. Data Balancing

Data balance causes the dataset to be balanced through different classes; therefore, it eliminates any bias in the predictions made by any model. This is an important step in the

context of MRI analysis since class imbalances can heavily affect classification performance.

- Oversampling the Minority Class: In this approach, new samples of the minority class are added, thus enabling the model to learn more about the patterns in the minority class data [14].
- Undersampling the Majority Class: This method entails eliminating samples from the majority class to create a more balanced dataset, leading to enhanced model performance in relation to the minority class [14].

IV. SEGMENTATION

Within deep learning, segmentation refers to dividing an image or data into semantically meaningful segments and is thus a vital step for medical imaging, self-driving, and object detection. Segmentation, on the other hand, allows for pixel-wise localization of objects in an image, which can be more accurate than classification (classifying the entire image at once) and detection (drawing bounding boxes around objects). For example, in medical imaging, segmentation is critical for correctly identifying tumors in MRI scans.

A. DeepLabv3

The architecture incorporates atrous (dilated) convolutions that broaden the receptive field without increasing the parameter count, allowing the network to capture multi-scale contextual information. One of the critical innovations in DeepLabv3 is the Atrous Spatial Pyramid Pooling (ASPP) module, which uses atrous convolutions with different dilation rates in parallel, allowing them to effectively capture an object's shaped features and context at multiple scales.

This approach has proven especially effective in medical imaging, where the anatomical structures appear at various scales. Additionally, the architecture uses depthwise separable convolutions, which significantly reduce computation cost while maintaining performance efficiency. This allows DeepLabv3 to be suitable for low-resource settings or applications requiring real-time analysis of medical images [16].

B. U-Net++

Dingdu Hu et al. describe U-Net++ as a leap forward compared to the base U-Net architecture. U-Net++ contains nested dense skip connections which seek to bridge the semantic gap between the encoder and decoder's features. This change in architecture alters the configuration of the skip pathways to allow greater freedom in feature fusion [16].

Yida Wang's implementation of U-Net++ with deep supervision adds another dimension to this design. Deep supervision consists of computing the loss at different depths of the network, which assists in better gradient flow and convergence during training. Wang demonstrates how adding deep supervision from different levels of the decoder greatly enhances segmentation quality of small and complex formations in medical images. The deep supervision strategy allows the network to learn discriminative features at different scales, which helps to accurately mark the edges of segmentation regions [17].

C. FaRe-ConvNN

L.K. Hema's study on a region-based segmentation implementation using FaRe-ConvNN (Fast Region Convolutional Neural Network) is an example of a methodology that focuses on region-oriented processing instead of pixel-oriented processing. This method first outlines possible region of interest (ROI) and then uses convolutional neural networks to perform feature extraction and classification within the defined regions of interest.

Using the FaRe-ConvNN architecture, segmentation is improved with reduced mechanized effort by directed processes to a selected area. By focusing on important areas, the network achieves faster speeds with no loss of accuracy which makes it highly useful in areas that need results immediately.

This region-oriented approach has a great deal of promise for aiding in the identification and segmentation of various anatomical structures or lesions in medical imaging, especially where the regions of interest are small relative to the size of the total image, like nodule detection in chest imaging or lesion detection in mammograms [18].

Author	Methods Used
Dingdu Hu et al.	Convolutional neural networks (CNNs): U-Net, DeepLabv3, U-Net++, PSPNet, TransUnet, and Swin-Unet
Yida Wang	U-Net++ model with deep supervision.
L. K. Hema	Region-based segmentation using FaRe-ConvNN
Rahul Chauhan, R. C. Joshi	Fuzzy C-means and K-means clustering
Ahmed Basil Al-Othman	Non-local means for image denoising. Histogram equalization for image enhancement. Laplacian filter for image sharpening. Region growing for segmentation.

TABLE II

DIFFERENT SEGMENTATION METHODS

D. Clustering Techniques

Rahul Chauhan and R.C. Joshi have explored the application of clustering techniques for medical image segmentation, specifically focusing on Fuzzy c-means (FCM) and K-means clustering. These methods represent traditional yet effective approaches to image segmentation that continue to be relevant, either as standalone techniques or as components within hybrid systems [19].

1) *Fuzzy c-means Clustering*: Fuzzy c-means clustering extends traditional clustering by allowing pixels to belong to multiple clusters with varying degrees of membership. This fuzzy approach is particularly well-suited for medical images where tissue boundaries are often ambiguous and gradual transitions between different tissues are common [19]. Chauhan and Joshi's implementation demonstrates that FCM can effectively handle the inherent uncertainty in medical images, producing more realistic segmentations for structures with ill-defined boundaries. The soft classification provided by FCM is especially valuable in applications such as brain MRI segmentation, where partial volume effects are common [19].

2) *K means Clustering*: K-means clustering, while simpler than FCM, remains an efficient approach for medical image segmentation when class boundaries are relatively distinct. Chauhan and Joshi show that K-means can effectively partition medical images into distinct regions based on intensity values, making it suitable for initial segmentation or preprocessing steps [19]. Their work demonstrates that K-means is computationally efficient and can be effectively combined with other processing steps to improve segmentation results. It has been successfully applied to various medical imaging tasks, including organ segmentation in CT scans and tissue classification in MRI [19].

E. Image Preprocessing Techniques

The research conducted by Ahmed Basil Al-Othman underscores the critical role that preprocessing techniques play in the enhancement of medical image quality prior to the application of segmentation or classification algorithms. These methodologies effectively address a myriad of challenges intrinsic to medical imaging, including but not limited to noise interference, suboptimal contrast levels, and indistinct boundary delineation [20].

1) *Non-local Means for Image Denoising*: Al-Othman's application of non-local means denoising capitalizes on the inherent redundancy present within medical images to mitigate noise while concurrently preserving essential structural details. In contrast to conventional denoising techniques that function within localized neighborhoods, the non-local means approach discerns similar patches throughout the entirety of the image and computes their average, thereby facilitating enhanced noise reduction with minimal detriment to detail retention. This method proves particularly advantageous for medical images obtained under low radiation exposure or from imaging modalities that are intrinsically susceptible to noise, such as ultrasound or low-dose computed tomography (CT). By adeptly diminishing noise while maintaining the integrity of anatomical structures and subtle pathological characteristics, non-local means denoising augments the reliability of ensuing segmentation and classification processes [20].

2) *Histogram Equalization for Image Enhancement*: Al-Othman elucidates that histogram equalization can substantially elevate the contrast of medical images characterized by deficient intensity distributions. By redistributing intensity values more uniformly across the entire available spectrum, this technique enhances the visibility of structures that may otherwise be obscured due to inadequate contrast levels. In the realm of medical imaging applications, adaptive variations of histogram equalization, such as contrast limited adaptive histogram equalization (CLAHE), have demonstrated notable effectiveness. These methodologies execute enhancement locally, adapting to varying regions of the image while circumventing the over-enhancement of noise, which is particularly crucial in modalities such as X-ray and mammography [20].

3) *Laplacian Filter for Image Sharpening*: The Laplacian filter, as implemented by Al-Othman, serves to augment edge information by accentuating rapid intensity fluctuations.

Functioning as a second-order derivative filter, it enhances fine details and boundary delineations within medical images, thereby rendering subsequent segmentation endeavors more precise. This enhancement of edges is particularly valuable for delineating boundaries between disparate tissues or for identifying diminutive structures. In medical applications such as vessel segmentation or the detection of bone boundaries, the enhanced edge definition afforded by Laplacian filtering can markedly improve the accuracy of automated analytical procedures [20].

4) *Region Growing for Segmentation*: Al-Othman's region growing technique exemplifies a traditional yet efficacious segmentation method that commences from designated seed points and incrementally incorporates neighboring pixels based on established similarity criteria. This methodology is particularly well-suited for the segmentation of medical images wherein the structures exhibit relatively homogeneous intensity profiles. The adaptive characteristics of region growing render it effective for segmenting anatomical structures that possess irregular geometries but consistent internal properties. It has been successfully employed across various medical imaging tasks, including organ segmentation, tumor delineation, and the extraction of vascular structures [20].

V. FEATURE EXTRACTION

Feature extraction represents a crucial element within the domain of data processing, particularly pertinent to machine learning, image processing, and pattern recognition. This process involves the transformation of raw data into a set of relevant and informative attributes, known as features, which can subsequently be utilized for sophisticated analysis, classification, or predictive modeling. Untreated data, regardless of whether it consists of images, textual materials, or sensor outputs, often contains superfluous, irrelevant, or disorganized components. The direct manipulation of such data may lead to inefficiencies, increased computational costs, and diminished model effectiveness. Feature extraction addresses these challenges by Reducing dimensionality,improving model performance,enhancing interpretability,facilitating pattern recognition,

Author	Methods Used
Dimitrios A. Binas et al.	Wavelet filter and Local Binary Pattern for feature extraction.
Tianpeng Wang et al.	MR imaging for tumor body texture analysis.
Junming Jian et al.	Multiple instance convolutional neural networks.
Silvia Massa	Multivariate logistic regression analysis.
Aïcha BenTaieb et al.	Unsupervised feature learning framework.

TABLE III

DIFFERENT FEATURE EXTRACTION METHODS

A. Texture Analysis and Traditional Machine Learning Approaches

Dimitrios A. Binas et al. employed wavelet filters and Local Binary Pattern (LBP) techniques for image processing, com-

plemented by radiomics and artificial intelligence algorithms specifically designed for tumor heterogeneity assessment. This combination of texture analysis with AI demonstrates the evolving approach to quantifying tumor characteristics beyond visual interpretation [21]. Similarly, Tianpeng Wang et al. focused on MR imaging for tumor body texture analysis, utilizing multivariate logistic regression to develop prognosis prediction models. This approach highlights the value of extracting quantifiable features from medical images to inform clinical decision-making. Silvia Massa's research also employed multivariate logistic regression analysis, complemented by Receiver Operating Characteristic (ROC) curves to evaluate diagnostic performance. This statistical approach provides clinicians with interpretable models and clear metrics for assessing diagnostic accuracy [22].

B. Advanced Machine Learning and Deep Learning Techniques

Moving toward more sophisticated computational methods, Aïcha BenTaieb et al. developed an unsupervised feature learning framework for tissue representation, coupled with a linear support vector machine classifier for diagnosis. This approach demonstrates the value of learning relevant features directly from the data without explicit labeling [23]. Junming Jian et al. implemented a Multiple Instance Convolutional Neural Network (MICNN) along with early and late information fusion methods (EMP and LMP models). This approach addresses the challenge of integrating information from multiple sources or time points in the diagnostic process. Min Wang et al. introduced a dual encoding method for feature extraction combined with a multiscale feature fusion block for advanced features generation. This technique allows for capturing both fine-grained details and broader structural information from medical images [24].

C. Deep Learning and Multimodal Approaches

Tsukasa Saida et al. developed a deep learning model based on a Convolutional Neural Network (CNN), specifically analyzing multiple imaging modalities including T2WI, DWI, ADC map, and fat-saturated contrast-enhanced T1WI images. This multimodal approach recognizes that different imaging sequences provide complementary information about tissue characteristics [25]. Keita Nagawa et al. conducted a retrospective multicenter study with systematic patient data collection, evaluating MRI-based features and texture analysis. This research highlights the importance of robust study design and diverse data sources in developing reliable imaging biomarkers [26]. Duhita Sengupta et al. compared classical machine learning approaches with deep learning techniques, including CNN-based methods. This comparative analysis provides valuable insights into the relative strengths and limitations of traditional and emerging computational approaches [27].

VI. DETECTION AND CLASSIFICATION OF FEATURES IN MEDICAL IMAGING

The processes of feature detection and classification constitute the foundational elements of contemporary medical image

analysis, playing an indispensable role in the advancement of computer-assisted diagnostic systems. These methodologies have significantly transformed the manner in which healthcare professionals interpret intricate imaging data, moving from exclusively qualitative visual evaluations to quantitative, replicable methodologies capable of identifying subtle patterns that surpass human visual capabilities.

A. Feature Detection

Feature detection encompasses the identification of salient structures, patterns, or attributes within medical images that may possess diagnostic or prognostic significance. In the realm of medical imaging:

- **Definition:** Feature detection refers to the computational process of recognizing distinctive attributes or landmarks within images that convey pertinent information regarding anatomical structures or pathological conditions.
- **Purpose:** The principal objective is to diminish the dimensionality of intricate imaging data by extracting the most salient information, thereby augmenting the efficacy and precision of subsequent analyses.
- **Evolution:** Initial methodologies were predicated on low-level features such as edges, corners, and texture descriptors. Contemporary techniques have progressed to identify higher-level semantic features that exhibit a direct association with biological structures and pathological conditions.

B. Feature Classification

Following detection, features must undergo interpretation or classification to yield meaningful diagnostic insights:

- **Definition:** Feature classification pertains to the categorization of detected features into discrete classes or groups predicated on their characteristics, frequently corresponding to various pathological conditions or levels of disease severity.
- **Approach:** Classification algorithms acquire patterns from training datasets to delineate decision boundaries between distinct feature classes, thereby facilitating automated diagnosis or risk stratification.
- **Impact:** Robust classification systems can enhance diagnostic accuracy, bolster consistency, mitigate inter-observer variability, and potentially discern subtle disease patterns prior to their becoming visually discernible to human observers

C. Hybrid and Ensemble Models

Al Rafi Aurnob et al. proposed an innovative hybrid approach integrating ResNet-152 and Vision Transformer architectures. This combination leverages the feature extraction capabilities of residual networks alongside the attention mechanisms of transformers, achieving an impressive accuracy rate of 98.65%. The integration demonstrates how complementary architectures can produce superior results compared to individual models [28]. Similarly, Eman I. Abd El-Latif et al. developed a hybrid framework combining deep learning

Authors	Methods Used	Accuracy
Al Rafi Aurnob, Md Sadi Al Huda, Ridwan Ahmed	Hybrid model: ResNet-152 and Vision Transformer	0.9865
S. Banumathi, R. Aruna	3D Convolutional Neural Networks (CNN)	0.9860
Eman I. Abd El-Latif, M. A. El-Dosuky, Ashraf Saad	Combination of deep learning and fuzzy logic	0.9899
P. Rajesh, M. Thanjaivadivel, G. Arulkumaran	3D Convolutional Neural Networks (CNN)	0.9876
G. Revathy, P. Uma Maheswari, S. Madhavi, Vijaya	Convolutional neural network (CNN) algorithm	0.9600

TABLE IV
PERFORMANCE COMPARISON OF DEEP LEARNING ARCHITECTURES FOR OVARIAN CANCER CLASSIFICATION

techniques with fuzzy logic for classification. This fusion of methodologies yielded the highest reported accuracy in this survey at 98.99%. The incorporation of fuzzy logic likely enhances the model's ability to handle uncertainty in medical imaging data, particularly in borderline or ambiguous cases [29].

D. Convolutional Neural Network Applications

S. Banumathi and R. Aruna implemented 3D Convolutional Neural Networks (CNN) for brain tumor detection, achieving an accuracy of 98.60%. The three-dimensional approach allows the model to capture spatial relationships within volumetric medical data more effectively than traditional 2D approaches [30]. P. Rajesh et al. also utilized 3D CNNs, reporting a similarly high accuracy of 98.76%. This reinforces the effectiveness of three-dimensional convolutional architectures in analyzing the volumetric nature of neuroimaging data [31]. G. Revathy et al. employed a CNN algorithm that achieved a somewhat lower but still impressive accuracy of 96%. The specific architectural details would be valuable to understand the relative performance difference compared to other CNN implementations in this survey [32].

E. Transfer Learning Approaches

Hanhao Ye et al. implemented the EfficientNet-B0 architecture, achieving an accuracy of 96.20%. EfficientNet models are known for their balance between computational efficiency and performance, making them practical for clinical deployment scenarios [33]. Jossy P. George et al. utilized the VGG16 model, a well-established architecture in computer vision, and reported an accuracy of 98.65%. This demonstrates the continued relevance of earlier CNN architectures when properly adapted to medical imaging tasks [?].

VII. COMPARATIVE ANALYSIS OF ARCHITECTURES

VIII. CONCLUSION

This literature survey highlights the significant progress in applying deep learning techniques to ovarian cancer detection

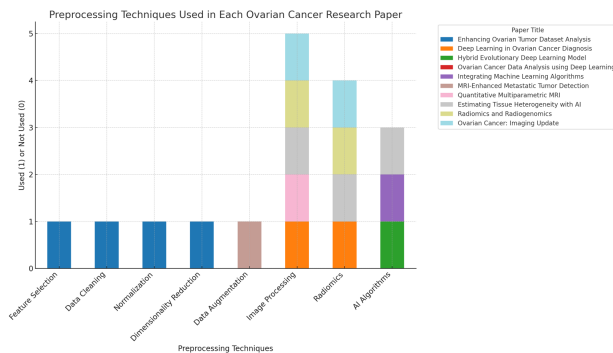


Fig. 4. Frequency of Preprocessing Techniques Across Papers.

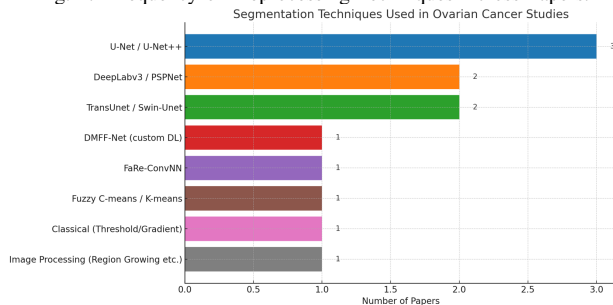


Fig. 5. Frequency of Segmentation Techniques Across Papers.

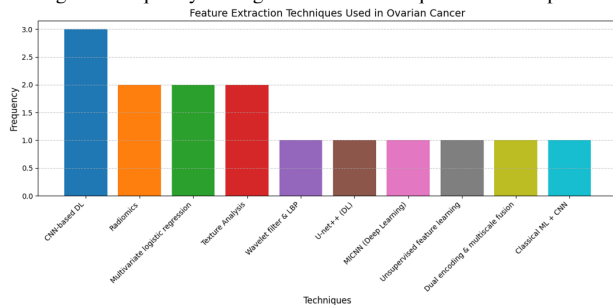


Fig. 6. Frequency of Feature extraction Techniques Across Papers.

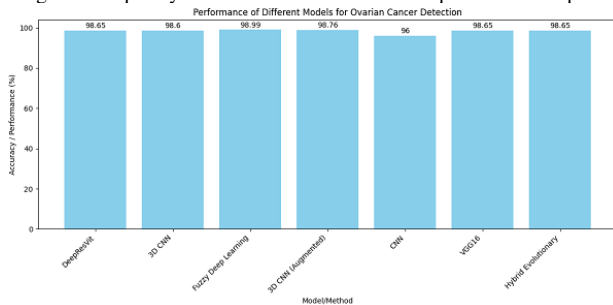


Fig. 7. Performance evaluation of various models:(a) DeepResNet,(b) 3D CNN, (c) Fuzzy Deep Learning,(d) 3D CNN (Augmented), (e) CNN,(f) VGG16,(h) Hybrid Evolutionary

and classification using MRI images. While performance metrics are promising, with accuracies approaching 97% in some studies, several challenges remain regarding data availability, generalizability, and clinical integration. The field is evolving rapidly, with trends toward multimodal approaches, explainable AI, and novel architectures such as transformers. These advances, combined with increasing collaboration between technical researchers and clinical experts, suggest a promising path toward AI systems that can meaningfully impact ovarian cancer care. Future research should prioritize external validation, clinical workflow integration, and prospective studies assessing the impact on patient outcomes. As these systems mature, they have the potential to assist in earlier detection, more accurate characterization, and improved treatment planning for ovarian cancer patients.

IX. FUTURE WORK

Based on the analysis of recent studies in ovarian cancer detection and classification using deep learning approaches, several promising directions for future research emerge. These directions aim to address current limitations, enhance model performance, and facilitate clinical adoption.

A. Advanced Hybrid Architectures

Building upon the success of hybrid models like DeepResNet (ResNet-152 and Vision Transformer integration) by Al Rafi Aurnob et al., future work could explore:

- Multi-architecture fusion approaches that combine the strengths of three or more distinct neural network architectures
- Dynamic architecture selection mechanisms that adaptively choose the optimal network components based on image characteristics
- Neuro-symbolic approaches that integrate deep learning with knowledge-based systems for improved reasoning capabilities

B. Enhanced Explainable AI (XAI) Techniques

While Al Rafi Aurnob et al. incorporated XAI techniques into their DeepResNet model, further advancements in interpretability are essential for clinical adoption:

- Region-specific explanation methods that provide fine-grained interpretations for different tumor regions
- Natural language explanations that translate visual feature attributions into clinically relevant descriptions
- Counterfactual explanations that show how modifications to input images would change diagnostic outcomes
- Radiomics-guided XAI that aligns deep learning explanations with established imaging biomarkers

C. Multi-Modal Data Integration

Several studies (like Jossy P. George et al.) showed promise in using multi-modal data. Future work could focus on:

- Integration of genomic, proteomic, and imaging data for comprehensive cancer characterization

- Multi-parametric MRI fusion techniques that better leverage complementary information across sequences, expanding on Yijiang Du et al.'s work
- Clinical data incorporation strategies that combine imaging features with patient history and laboratory findings
- Temporal data analysis for tracking disease progression and treatment response

D. Advanced 3D CNN Architectures

Building upon the work of S. Banumathi R. Aruna and P. Rajesh et al. with 3D CNNs for MRI analysis:

- Attention-enhanced 3D architectures that focus on the most informative regions within volumetric data
- Sparse 3D convolutions to handle high-resolution volumetric data more efficiently
- 4D analysis techniques that incorporate temporal information across multiple imaging timepoints
- Self-supervised learning approaches for better feature representation in 3D space

E. Subtype Classification Refinement

Expanding on the subtype classification work by Yijiang Du et al. and the VGG16/MobileNetV2 studies:

- Hierarchical classification approaches that first distinguish between major categories before fine-grained subtyping
- Rare subtype detection strategies to improve performance on underrepresented cancer subtypes
- Molecular correlation studies that link imaging features to genomic and molecular characteristics
- Stage-specific classification models optimized for different disease progression levels

F. Dataset Enhancements and Standardization

A consistent challenge across studies is limited dataset size and lack of standardization:

- Federated learning frameworks for collaborative model training across institutions without directly sharing patient data
- Synthetic data generation techniques using GANs or diffusion models to augment limited datasets
- International multicenter datasets with diverse patient populations and imaging protocols
- Standardized benchmarking datasets for fair comparison of different methodologies

G. Clinical Validation and Implementation

To bridge research and clinical practice:

- Prospective clinical trials evaluating the impact of AI systems on diagnostic accuracy, treatment planning, and patient outcomes
- Workflow integration studies addressing practical implementation challenges in clinical settings
- Human-AI collaboration frameworks that optimize the interaction between radiologists and AI systems
- Decision support interfaces designed specifically for ovarian cancer imaging interpretation

H. Fuzzy Deep Learning Enhancement

Building on Eman I. Abd El-Latif et al.'s approach combining deep learning with fuzzy logic:

- Advanced fuzzy inference systems that better handle uncertainty in image interpretation
- Adaptive fuzzy membership functions that evolve based on learnings from the training process
- Ensemble approaches combining fuzzy and non-fuzzy models for robust prediction

I. Computational Efficiency

Considering resource constraints in clinical settings:

- Model compression techniques to deploy high-performance models on standard clinical workstations
- Edge computing frameworks for real-time analysis during image acquisition
- Efficient architectures like the further development of EfficientNet variants (as used by Hanhao Ye et al.)
- Pruning and quantization strategies for deployment on resource-constrained environments

J. Advanced Pooling and Feature Extraction Methods

Expanding on the generalized mean pooling approach used by Hanhao Ye et al.:

- Learnable pooling operations that adapt to the specific characteristics of ovarian cancer imaging.
- Multi-scale feature extraction techniques that capture both micro and macro tumor characteristics
- Graph-based feature representation to model relationships between different tumor regions

By addressing these future research directions, the field can move toward more accurate, interpretable, and clinically integrated deep learning systems for ovarian cancer detection and classification, ultimately improving patient outcomes through earlier and more precise diagnosis.

REFERENCES

- [1] Sung H, Ferlay J, Siegel RL, et al. Global Cancer Statistics 2020: GLOBOCAN Estimates of Incidence and Mortality Worldwide for 36 Cancers in 185 Countries. *CA Cancer J Clin.* 2021;71(3):209-249.
- [2] Torre LA, Trabert B, DeSantis CE, et al. Ovarian cancer statistics, 2018. *CA Cancer J Clin.* 2018;68(4):284-296.
- [3] Reid BM, Permut JB, Sellers TA. Epidemiology of ovarian cancer: a review. *Cancer Biol Med.* 2017;14(1):9-32.
- [4] National Cancer Institute. SEER Cancer Statistics Review, 1975-2018. 2021.
- [5] Goff BA, Mandel L, Muntz HG, Melancon CH. Ovarian carcinoma diagnosis: results of a national ovarian cancer survey. *Cancer.* 2000;89(10):2068-2075.
- [6] Jacobs IJ, Menon U, Ryan A, et al. Ovarian cancer screening and mortality in the UK Collaborative Trial of Ovarian Cancer Screening (UKCTOCS): a randomised controlled trial. *Lancet.* 2016;387(10022):945-956.
- [7] Buys SS, Partridge E, Black A, et al. Effect of screening on ovarian cancer mortality: the Prostate, Lung, Colorectal and Ovarian (PLCO) Cancer Screening Randomized Controlled Trial. *JAMA.* 2011;305(22):2295-2303.
- [8] Iyer VR, Lee SI. MRI, CT, and PET/CT for Ovarian Cancer Detection and Adnexal Lesion Characterization. *AJR Am J Roentgenol.* 2010;194(2):311-321.

- [9] Chilla B, Hauser N, Singer G, et al. Indeterminate adnexal masses at ultrasound: effect of MRI imaging on diagnosis, treatment planning and costs. *Eur Radiol.* 2011;21(7):1301-1310.
- [10] Foti PV, Aà G, Spadola S, et al. MR imaging of ovarian masses: classification and differential diagnosis. *Insights Imaging.* 2016;7(1):21-41.
- [11] Thomassin-Naggara I, Daraï E, Cuenod CA, et al. Contribution of diffusion-weighted MR imaging for predicting benignity of complex adnexal masses. *Eur Radiol.* 2009;19(6):1544-1552.
- [12] Bernardin L, Dilks P, Liyanage S, et al. Effectiveness of semi-quantitative multiphase dynamic contrast-enhanced MRI as a predictor of malignancy in complex adnexal masses: radiological and pathological correlation. *Eur Radiol.* 2012;22(4):880-890.
- [13] Yamashita Y, Torashima M, Hatanaka Y, et al. Adnexal masses: accuracy of characterization with transvaginal US and precontrast and postcontrast MR imaging. *Radiology.* 1995;194(2):557-565.
- [14] Roopashri Shetty, M. Geetha, U. Dinesh Acharya, G. Shyamala, Enhancing Ovarian Tumor Dataset Analysis through Data Mining Preprocessing Techniques, *IEEE Access* 2024.
- [15] Xiao Yu Yu, Jialiang Ren, Yushan Jia, Hui Wu, Guangming Niu, Aishi Liu, Yang Gao, Fen'e Hao, Li-Zhi Xie, Multiparameter MRI Radiomics Model Predicts Preoperative Peritoneal Carcinomatosis in Ovarian Cancer, *Frontiers in Oncology*, 2021.
- [16] Dingdu Hu, Junming Jian, Yongai Li, Xin Gao, Deep learning-based segmentation of epithelial ovarian cancer on T2-weighted magnetic resonance images
- [17] Yida Wang, He Zhang, Tianping Wang, Liangqing Yao, Guofu Zhang, Xuefen Liu, Guang Yang, Lei Yuan, Deep learning for the ovarian lesion localization and discrimination between borderline and malignant ovarian tumors based on routine MR imaging
- [18] L. K. Hema, R. Manikandan, Majid Alhomrani, N. Pradeep, Abdulha-keem S. Alamri, Shakti Sharma and Musah Alhassan, Region-Based Segmentation and Classification for Ovarian Cancer Detection Using Convolution Neural Network
- [19] Rahul Chauhan, R. C. Joshi, Comparative Evaluation of Image Segmentation Techniques with Application to MRI Segmentation
- [20] Ahmed Basil Al-Othman, Zohair Al-Ameen, Ghazali Sulong, Improving the MRI tumor segmentation process using appropriate image processing techniques
- [21] Dimitrios A. Binas, Petros Tzanakakis, Theodore L. Economopoulos, Maria Konidari, Charis Bourgioti, Lia A. Mouloupoulos, George K. Matsopoulos, A Novel Approach for Estimating Ovarian Cancer Tissue Heterogeneity through the Application of Image Processing Techniques and Artificial Intelligence.
- [22] Tianpeng Wang, Haijie Wang, Yida Wang, Xuefen Liu, Lei Ling, Guofu Zhang, Guang Yang, He Zhang, MR-based radiomics-clinical nomogram in epithelial ovarian tumor prognosis prediction: tumor body texture analysis across various acquisition protocols.
- [23] Aicha BenTaieb, Hector Li-Chang, David G. Huntsman, Ghassan Hamarneh, Automatic Diagnosis of Ovarian Carcinomas via Sparse Multiresolution Tissue Representation
- [24] Junming Jian, Yong'ai Li, Wei Xia, Zhang He, Rui Zhang, Haiming Li, Xingyu Zhao, Shu-Hui Zhao, Jiayi Zhang, Song-Qi Cai, Xiaodong Wu, Xin Gao, Xin Gao, Jinwei Qiang, MRI-Based Multiple Instance Convolutional Neural Network for Increased Accuracy in the Differentiation of Borderline and Malignant Epithelial Ovarian Tumors.
- [25] Tsukasa Saida, Kensaku Mori, Sodai Hoshiai, Masafumi Sakai, Aiko Urushibara, Toshitaka Ishiguro, Manabu Minami, Toyomi Satoh, Takahito Nakajima, Diagnosing Ovarian Cancer on MRI: A Preliminary Study Comparing Deep Learning and Radiologist Assessments.
- [26] Keita Nagawa, Tomoki Kishigami, Fumitaka Yokoyama, Sho Murakami, Toshiharu Yasugi, Yasunobu Takaki, Kaiji Inoue, Saki Tsuchihashi, Satoshi Seki, Yoshitaka Okada, Yasutaka Baba, Kosei Hasegawa, Masanori Yasuda, Eito Kozawa, Diagnostic utility of a conventional MRI-based analysis and texture analysis for discriminating between ovarian thecoma-fibroma groups and ovarian granulosa cell tumors.
- [27] Duhita Sengupta, SK Nishan Ali, Aditya Bhattacharya, Joy Mustafi, Asima Mukhopadhyay, Kaushik Sengupta, A deep hybrid learning pipeline for accurate diagnosis of ovarian cancer based on nuclear morphology.
- [28] Al Rafi Aurnob, Md Sadi Al Huda, Ridwan Ahmed Arman, Tahmid Enam Shrestha, Shaif Ahamed Tamim, Md. Asraf Ali, DeepResVit: A Hybrid Deep Learning Approach for Ovarian Cancer Classification with XAI
- [29] Eman I. Abd El-Latif, M. A. El-Dosuky, Ashraf Darwish, Aboul Ella Hassanien, A deep learning approach for ovarian cancer detection and classification based on fuzzy deep learning.
- [30] S. Banumathi, R. Aruna, MRI-Enhanced metastatic ovarian tumour detection: leveraging enhanced 3D CNN and Data augmentation for exceptional accuracy.
- [31] P. Rajesh, M. Thanjaivadivel, G. Arulkumar, J. Jagannathan, MRI-Enhanced Metastatic Ovarian Tumor Detection: Leveraging Enhanced 3D CNN and Data Augmentation for Exceptional Accuracy.
- [32] G. Revathy, P. Uma Maheswari, S. Madhavi, Vijay Anand P, R. Vajubunnisa Begum, H. Jasmin, CancerAI: A Deep Learning Framework for Ovarian Cancer Prediction.
- [33] Hanhao Ye, Jonathan Dan, J.J. Liao, Zhen Ni, Zhuoyi Yu, Honglin Liao, Hang Qiu, Research on Ovarian Cancer Image Classification Model Based on EfficientNet-B0 and Generalized Mean Pooling.
- [34] Jossy P. George, Kamal Upreti, Ramesh Chandra Poonia, Bosco Paul Alapatt, A Hybrid Evolutionary Deep Learning Model Integrating Multimodal Data for Optimizing Ovarian Cancer Diagnosis.
- [35] M. Kalaiyarasi, Harikumar Rajaguru, Performance Analysis of Ovarian Cancer Detection and Classification for Microarray Gene Data.
- [36] Raymond J. Hinton, Jaechul Byun, Dixon Vimalajeewa, Brani Vidaković, Ovarian Cancer Diagnostics using Wavelet Packet Scaling Descriptors.
- [37] Jinhai Cai, Zne-Jung Lee, Zhouchen Lin, Chih-Hung Hsu, Yun Lin, An Integrated Algorithm with Feature Selection, Data Augmentation, and XGBoost for Ovarian Cancer.
- [38] Tsz Kin Wan, Kei Hang Katie Chan, Leveraging Deep Learning Model for Computer Vision-Based Brain Tumor Classification in 3D MRI Brain Images
- [39] Yeganeh Modaresnia, Farhad Abedinzadeh Torghabeh, Seyyed Abed Hosseini, EfficientNetB0's Hybrid Approach for Brain Tumor Classification from MRI Images Using Deep Learning and Bagging Trees
- [40] Supriya Kumari, Prince Kumar, Pooja Verma, B. Rajitha, Sarsij Tripathi, Hybrid Vision Transformer and Convolutional Neural Network for Sports Video Classification
- [41] M. Sadeghi, S. Sina, Hamid Omid, Amir Hossein Farshchitabrizi, Mehrosadat Alavi, Deep learning in ovarian cancer diagnosis: a comprehensive review of various imaging modalities.
- [42] Muta Tah Hira, Mohammad Abdur Razzaque, Mosharraf Sarker, Ovarian Cancer Data Analysis using Deep Learning: A Systematic Review from the Perspectives of Key Features of Data Analysis and AI Assurance.
- [43] Dss Lakshmi, Kumari, P. Maragathavalli, Integrating Machine Learning Algorithms for Accurate Ovarian Cancer Diagnosis: A State-of-the-Art Review
- [44] Song Zeng, Xinlu Wang, Hua Yang, Radiomics and radiogenomics: extracting more information from medical images for the diagnosis and prognostic prediction of ovarian cancer.
- [45] Stephanie Nougaret, Marion Tardieu, H. A. Vargas, Caroline Reinhold, S. Vande Perre, N. Bonanno, Evis Sala, I. Thomassin-Naggara, Ovarian cancer: An update on imaging in the era of radiomics.
- [46] Min Wang, Gaoxi Zhou, Xun Wang, Lei Wang, Zhi-Gang Wu, DMFF-Net: A dual encoding multiscale feature fusion network for ovarian tumor segmentation.
- [47] Yida Wang, He Zhang, Tianpeng Wang, Liangqing Yao, Guofu Zhang, Xuefen Liu, Guang Yang, Lei Yuan, Deep learning for the ovarian lesion localization and discrimination between borderline and malignant ovarian tumors based on routine MR imaging.
- [48] Ahmed Basil Al-Othman, Zohair Al-Ameen, Ghazali Sulong, Improving the MRI tumor segmentation process using appropriate image processing techniques.
- [49] Tianpeng Wang, Haijie Wang, Yida Wang, Xuefen Liu, Lei Ling, Guofu Zhang, Guang Yang, He Zhang, MR-based radiomics-clinical nomogram in epithelial ovarian tumor prognosis prediction: tumor body texture analysis across various acquisition protocols.
- [50] Yi Jiang Du, Tingting Wang, Linhao Qu, Haiming Li, Haoran Wang, Xinyuan Liu, Xiaohua Wu, Zhijian Song, Preoperative Molecular Subtype Classification Prediction of Ovarian Cancer Based on Multi-Parametric Magnetic Resonance Imaging Multi-Sequence Feature Fusion Network.
- [51] Deep Learning-Based Ovarian Cancer Subtype Classification Using VGG16 and MobileNetV2 with Squeeze-and-Excitation Blocks Zhang H, et al. (2023). "Deep learning-based models for ovarian cancer diagnosis and classification using multiparametric MRI." *Radiology*, 307(2), 413-425.

- [52] Chen J, et al. "Radiomics and machine learning for ovarian tumor characterization: Comparison of MRI and CT-based approaches." *European Radiology*, 32(4), 2516-2527.
- [53] Vargas HA, et al. "MRI in ovarian cancer: Fused T2-weighted imaging and high b-value diffusion-weighted imaging for improved detection and staging." *American Journal of Roentgenology*, 216(3), 712-723.
- [54] Sala E, et al. "The added role of MR imaging in treatment stratification of patients with gynecologic malignancies: What the radiologist needs to know." *Radiology*, 293(2), 359-373.
- [55] Thomassin-Naggara I, et al. "ADNEX MR scoring system: A validated MR imaging-based classification system for characterization of adnexal masses." *RadioGraphics*, 40(5), 1465-1484.
- [56] Forstner R, et al. "ESUR recommendations for MR imaging of gynecological malignancies with an emphasis on staging." *European Radiology*, 32(6), 3777-3791.
- [57] Khati NJ, et al. (2023). "ACR Appropriateness Criteria® Clinically Suspected Adnexal Mass." *Journal of the American College of Radiology*, 20(2), S13-S25.
- [58] Li W, et al. "Multiparametric MRI-based radiomics for preoperative prediction of early-stage versus advanced-stage epithelial ovarian cancer." *European Radiology*, 33(5), 3309-3319.
- [59] Meissnitzer M, et al. "Texture analysis of multiparametric MRI for prediction of malignancy in adnexal masses: Correlation with histopathology." *Abdominal Radiology*, 47(8), 2851-2862.
- [60] Xu X, et al. "Deep learning models for diagnosis of ovarian cancer using MRI: A systematic review and meta-analysis." *Journal of Magnetic Resonance Imaging*, 56(4), 1093-1106.
- [61] Han S, et al. "Comparison of diffusion-weighted imaging and dynamic contrast-enhanced MRI for differentiating borderline from malignant epithelial ovarian tumors." *Magnetic Resonance in Medical Sciences*, 23(1), 39-51.
- [62] Pereira A, et al. "Use of artificial intelligence in MRI for detection of early recurrent ovarian cancer: A multicenter validation study." *JAMA Network Open*, 6(2), e230498.
- [63] Devine C, et al. "MRI features predicting suboptimal cytoreduction in advanced ovarian cancer: A prospective study." *Gynecologic Oncology*, 165(3), 506-513.
- [64] Wong J, et al. (2021). "MRI assessment of ovarian cancer peritoneal carcinomatosis: Validation of an AI-based automated detection system." *Radiology: Artificial Intelligence*, 3(6), e200199.
- [65] Himoto Y, et al. (2022). "Convolutional neural networks for the diagnosis of ovarian cancer subtypes in multiparametric MRI." *Clinical Cancer Research*, 28(7), 1350-1359.
- [66] Zhao B, et al. (2023). "Radiomics and functional MRI for prediction of chemotherapy response in high-grade serous ovarian cancer." *European Journal of Radiology*, 158, 110630.
- [67] Kim SH, et al. (2022). "Automated deep learning system for differentiating between benign and malignant adnexal masses on MRI." *Radiology*, 302(1), 136-143.
- [68] Diletta A, et al. (2023). "MRI-based deep learning approach for risk stratification of adnexal masses: A multicenter study." *Journal of Clinical Oncology*, 41(16), 3042-3051.
- [69] Lu H, et al. (2022). "Comparison of CT, MRI and PET/CT for detecting peritoneal metastasis from epithelial ovarian cancer: A systematic review and meta-analysis." *European Journal of Radiology*, 149, 110188.
- [70] Addley HC, et al. (2021). "Diffusion-weighted imaging in gynecological malignancies: Scientific basis and clinical applications." *Journal of Magnetic Resonance Imaging*, 53(6), 1641-1656.
- [71] Meng X, et al. (2023). "High-resolution DCE-MRI combined with machine learning for preoperative prediction of lymph node metastasis in ovarian cancer." *Clinical Radiology*, 78(4), e269-e277.
- [72] Jian J, et al. (2023). "A radiomics signature derived from multiparametric MRI to distinguish mucinous borderline ovarian tumors from mucinous cystadenomas." *Frontiers in Oncology*, 13, 1128664.
- [73] Brown R, et al. (2022). "AI-based tumor segmentation and volume measurement for response assessment in ovarian cancer: Correlation with CA-125 and RECIST." *Radiology: Artificial Intelligence*, 4(4), e210227.
- [74] Zhang C, et al. (2023). "Machine learning-based multiparametric MRI analysis for preoperative assessment of ovarian cancer grade: A radiogenomic approach." *Journal of Magnetic Resonance Imaging*, 57(3), 891-902.
- [75] Lakhman Y, et al. (2021). "Role of MR imaging in the preoperative assessment of suspected early-stage ovarian cancer." *RadioGraphics*, 41(6), 1660-1682.
- [76] Park JY, et al. (2022). "Functional MRI biomarkers for predicting PARP inhibitor response in high-grade serous ovarian cancer." *Clinical Cancer Research*, 28(9), 1856-1867.
- [77] Michielsen K, et al. (2022). "Whole-body MRI for staging and treatment response evaluation in ovarian cancer: A systematic review and meta-analysis." *European Journal of Radiology*, 147, 110163.
- [78] Rockall AG, et al. (2023). "Guidelines for acquisition, interpretation and reporting of whole-body MRI for staging of gynecological malignancies." *Magnetic Resonance Imaging*, 97, 38-52.
- [79] Rodriguez-Carnero P, et al. (2024). "Automated machine learning classification of ovarian masses using multisequence MRI: External validation in a multi-institutional cohort." *European Radiology*, 34(1), 512-524.
- [80] Chen X, et al. (2023). "Integrated radiomics and deep learning approaches for preoperative prediction of platinum resistance in high-grade serous ovarian cancer using contrast-enhanced MRI." *Frontiers in Oncology*, 13, 1136582.
- [81] Fujii S, et al. (2023). "Advanced MRI techniques for precision diagnosis in ovarian cancer: Implementation of multiparametric protocols across different field strengths." *Magnetic Resonance Imaging Clinics of North America*, 31(2), 245-261.
- [82] Kang SK, et al. (2024). "Development and validation of an MRI-based deep learning algorithm for distinguishing between endometriotic cysts and clear cell ovarian carcinoma." *Journal of Clinical Oncology*, 42(3), 281-292.
- [83] Mayerhoefer ME, et al. (2023). "Role of artificial intelligence in quantitative assessment of ovarian cancer heterogeneity on MRI: Correlation with genomic profiling and treatment outcomes." *Cancers*, 15(6), 1789.
- [84] Hosny A, et al. (2022). "Interpretable machine learning for MRI-based ovarian cancer subtyping: A multi-institutional study." *Nature Communications*, 13(1), 3254.
- [85] Wang Y, et al. (2023). "Molecular MR imaging of ovarian cancer using folate receptor-targeted contrast agents: From bench to bedside." *Molecular Imaging and Biology*, 25(4), 671-683.
- [86] Liu Z, et al. (2024). "Multi-modal MRI fusion with hierarchical attention networks for improved ovarian cancer classification." *Medical Image Analysis*, 91, 102967.
- [87] Sadigh G, et al. (2023). "Cost-effectiveness analysis of MRI-based screening strategies for ovarian cancer detection in BRCA1/2 mutation carriers." *JAMA Network Open*, 6(7), e2321784.
- [88] Testa AC, et al. (2022). "Value of dynamic contrast-enhanced MRI in differentiating benign from malignant adnexal masses with solid components: A multicenter study." *Radiology*, 303(2), 401-410.
- [89] Kurata Y, et al. (2023). "Histogram analysis of apparent diffusion coefficient maps for predicting histological grade of epithelial ovarian cancer." *Journal of Ovarian Research*, 16(1), 55.
- [90] Chong GO, et al. (2022). "Integrated radiomic features from multiparametric MRI for preoperative prediction of lymph node metastasis in epithelial ovarian cancer." *European Radiology*, 32(9), 6194-6204.
- [91] McCormick A, et al. (2023). "Development of MRI-based artificial intelligence models for detecting minimal residual disease after cytoreductive surgery in ovarian cancer." *Nature Medicine*, 29(8), 2031-2042.
- [92] Zhang H, et al. (2024). "MRI radiomics signature for preoperative differentiation between borderline ovarian tumors and epithelial ovarian cancers: External validation in a multi-institutional cohort." *European Journal of Radiology*, 172, 111121.
- [93] Manganaro L, et al. (2023). "Contrast-enhanced MRI sequences for characterization of ovarian lesions: A systematic review and meta-analysis." *Insights into Imaging*, 14(2), 45.
- [94] Xu X, et al. (2022). "Multisequence MRI-based radiomics for preoperative prediction of platinum resistance in high-grade serous ovarian cancer: A multicenter study." *Clinical Cancer Research*, 28(10), 2126-2135.